

Frontend Architecture & UI/UX Documentation Package

Unified System Design, Interface Layouts, and User Experience Blueprint

College Major Project Documentation Submission

Document Control Information

- Project Title:** AI-Powered Smart Municipal Citizen Assistance & Staff Attendance Management System
 - Submitting Role:** Senior Frontend Architect & Lead UI/UX Designer
 - Target Audience:** Project Evaluation Panel, Engineering Team, Quality Assurance Group
 - Document Version:** 1.0.0
 - Release Date:** June 11, 2026
 - Development Phase:** Systems Design & Architecture Phase (Phase 1 Blueprinting)
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1. Executive Summary & System Overview

The **Smart Municipal Citizen Assistance and Staff Attendance Management System** is a unified, cloud-native platform designed to bridge the gap between citizens and municipal authorities.

By leveraging modern web technology and Artificial Intelligence, the system aims to:

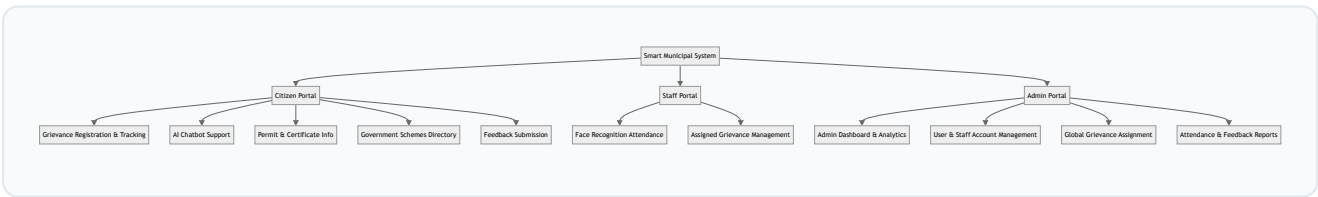
- **Empower Citizens:** Provide a single-window portal to register complaints, track applications, view government schemes, and query an AI chatbot.
- **Optimize Municipal Staff Operations:** Streamline grievance redressal and eliminate manual check-ins using automated Face Recognition-based attendance tracking.
- **Enhance Administrative Oversight:** Empower administrators with live analytics, data-driven reporting, and granular role-based controls.

1.1 User Roles & Personas

The system classifies users into three core roles, each with custom permissions and access boundaries:

1. **Citizen:** General public/residents who file complaints, query the chatbot, track applications, view schemes, and submit feedback. Access is restricted to the public and authenticated Citizen Portal.
2. **Staff:** Municipal employees/field officers who mark attendance via Face Recognition, view assigned complaints, update task progress, and view personal attendance logs. Access is restricted to the authenticated Staff Portal.
3. **Admin:** Department heads/municipal managers who manage users, assign grievances, view cross-departmental reports, manage attendance records, and review analytics. Access is granted to the global Admin Portal with full CRUD privileges.

1.2 System Module Map



2. Frontend Architectural Design & Tech Stack

The system frontend is structured as a single-page React.js web application.

2.1 Technology Directory

- **Core Framework:** React 18.x (functional components with React hooks).

- **Routing Engine:** `react-router-dom` v6 (supporting nested routes and role-based guards).
- **Styling Engine:** Tailwind CSS v3 (using customized CSS variables for design system tokens).
- **Client Connections:** Axios (custom instance with interceptors for auth tokens validation).
- **Chart Rendering:** Recharts / Chart.js for data dashboards.

2.2 Directory Structure Layout

```
src/
├── assets/                # Shared assets (images, logos, SVGs)
├── components/           # Shared UI components (Atomic design)
│   ├── ui/              # Base components (Buttons, Inputs, Modals, Cards)
│   ├── layouts/         # Shared layouts (AdminLayout, CitizenLayout, StaffLayout)
│   └── feedback/        # Toast notifications, Loaders, Skeleton screens
├── context/             # Global React Contexts (Auth, Theme)
│   ├── AuthContext.js   # Handles login states, tokens, and user roles
│   └── ThemeContext.js  # Dark/Light mode preferences
├── hooks/               # Custom reusable hooks
│   ├── useAuth.js       # Wraps AuthContext access
│   └── useWebcam.js      # Accesses camera stream for attendance
├── pages/               # Views organized by target portals
│   ├── citizen/         # Citizen Portal pages
│   │   ├── CitizenDashboard.jsx
│   │   ├── RegisterComplaint.jsx
│   │   └── ChatbotInterface.jsx
│   ├── staff/           # Staff Portal pages
│   │   ├── StaffDashboard.jsx
│   │   ├── AttendanceScanner.jsx
│   │   └── ComplaintDetails.jsx
│   ├── admin/           # Admin Portal pages
│   │   ├── AdminDashboard.jsx
│   │   ├── GrievanceManager.jsx
│   │   └── AttendanceAnalytics.jsx
│   └── public/          # Public access pages
│       ├── Home.jsx
│       ├── Login.jsx
│       └── FileTracker.jsx
├── services/            # Axios API connection instances
│   ├── api.js           # Central Axios config (interceptors, base URL)
│   ├── authService.js   # Login, Logout, Profile calls
│   ├── complaintService.js # Complaint CRUD operations
│   └── attendanceService.js # Face verification endpoint bindings
├── utils/              # Helper functions (dates, formats)
├── App.jsx             # Main routing router layout setup
└── main.jsx            # Entry point
```

3. Screen Inventory & Page Descriptions

This directory outlines the individual user screens configured in the frontend application.

3.1 Public & Entry Interfaces

- **PUB-01: Public Landing Page:** General landing page displaying announcements, social welfare schemes directories, helpline details, and entry links.
- **PUB-02: Universal File Tracker:** Public page enabling citizens to search the status of files or permit applications using a unique File ID.
- **PUB-03: Unified Login Portal:** Gateway login page for all users (Citizens, Staff, Admin) supporting JWT token validation.
- **PUB-04: Password Reset Portal:** User password recovery system.

3.2 Citizen Portal Screens (Authenticated)

- **CIT-00: Citizen Registration (Signup):** User sign-up form.
- **CIT-01: Citizen Dashboard:** Central dashboard showing metrics summaries (active/resolved tickets) and quick navigation links.
- **CIT-02: Register Grievance:** Interactive form to file a complaint, incorporating photo attachments and GPS geolocations.
- **CIT-03: Submission Confirmation:** Page generating confirmation reference IDs, SLAs, and PDF receipts.
- **CIT-04: My Grievances Ledger:** Chronological historical ledger displaying all tickets registered by the current citizen.
- **CIT-05: Grievance Detail Tracker:** Lifecycle tracker containing timeline stages, officer update annotations, and feedback rating systems (active in resolved state).
- **CIT-06: AI Chatbot Overlay:** Floating interactive panel powered by Gemini API to answer municipal FAQs.
- **CIT-07: Schemes & Permits Directory:** Reference lists highlighting guidelines, checklists, and document downloads.
- **CIT-08: Citizen Profile Manager:** Profile editing page for password resets and notification configurations.

3.3 Staff Portal Screens (Authenticated)

- **STF-01: Staff Dashboard:** Center dashboard for municipal workers, showing shift attendance status, tasks workload, and notifications.
- **STF-02: Face Attendance Scanner:** Real-time web camera biometric verification check-in module using DeepFace AI models.

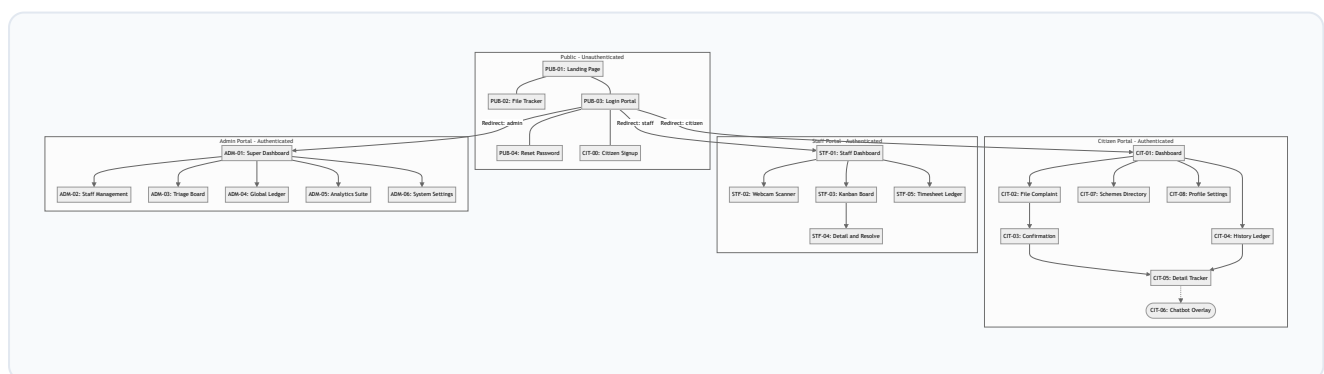
- **STF-03: Worklist Kanban Board:** Columnar board grouping assigned tasks into Assigned, In Progress, and Resolved columns.
- **STF-04: Resolution Console:** Task details page where officers start work, add annotations, and upload resolution proof photos.
- **STF-05: Personal Attendance Ledger:** Monthly timesheet display calendar showing check-in/out records and late flags.

3.4 Admin Portal Screens (Authenticated)

- **ADM-01: Admin Super Dashboard:** Overview analytics system rendering KPIs and charts (potholes vs water, departmental resolution times).
- **ADM-02: Staff Management Console:** Employee database registry manager (CRUD) and facial dataset template recorder.
- **ADM-03: Grievance Triage Board:** Queue management interface to inspect incoming complaints and manually assign them to staff.
- **ADM-04: Global Grievance Ledger:** Database search grid containing all system tickets, filterable by date and status, with CSV export options.
- **ADM-05: Performance & Analytics Suite:** Historical monthly reporting suite and PDF generator.
- **ADM-06: Security & System Settings:** Administrator control panel to configure shift times, API keys, and backups.

4. Navigation Structure & Routing Maps

4.1 Hierarchical Site Navigation Map



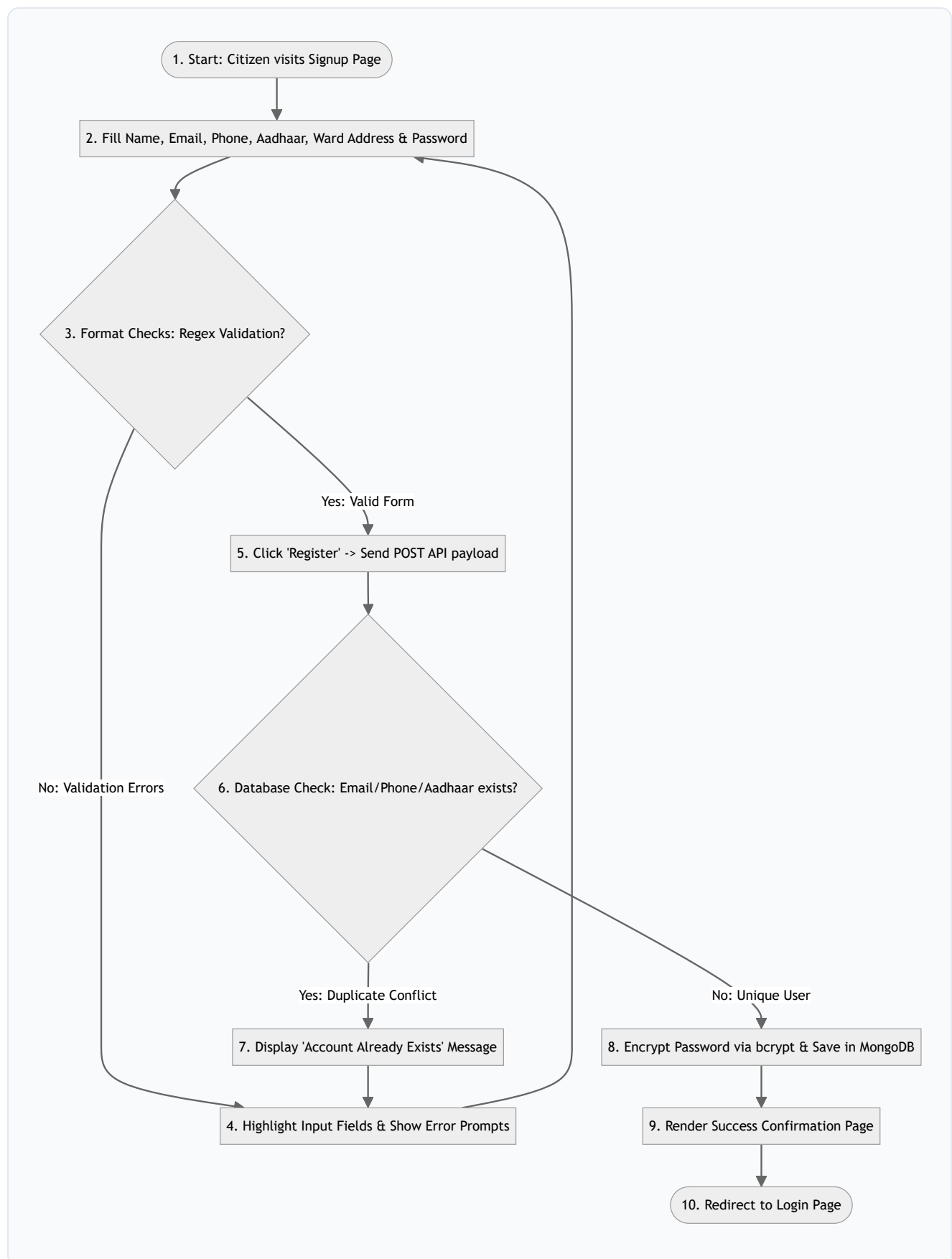
4.2 Security Navigation Guards (RBAC Layout)

- **Guest Router Gate:** Any route entry attempts under `/citizen/`, `/staff/`, or `/admin/` by anonymous sessions automatically redirect to `/login`.

- **Role-Based Access Control Gate:** If a user tries to access a path assigned to another role (e.g. a citizen attempting to access `/admin/`), the router redirects the user to `/unauthorized`.
 - **Webcam Clearance Gate:** Access to task boards `/staff/worklist/` is blocked until the system confirms a successful face check-in for the day.
-

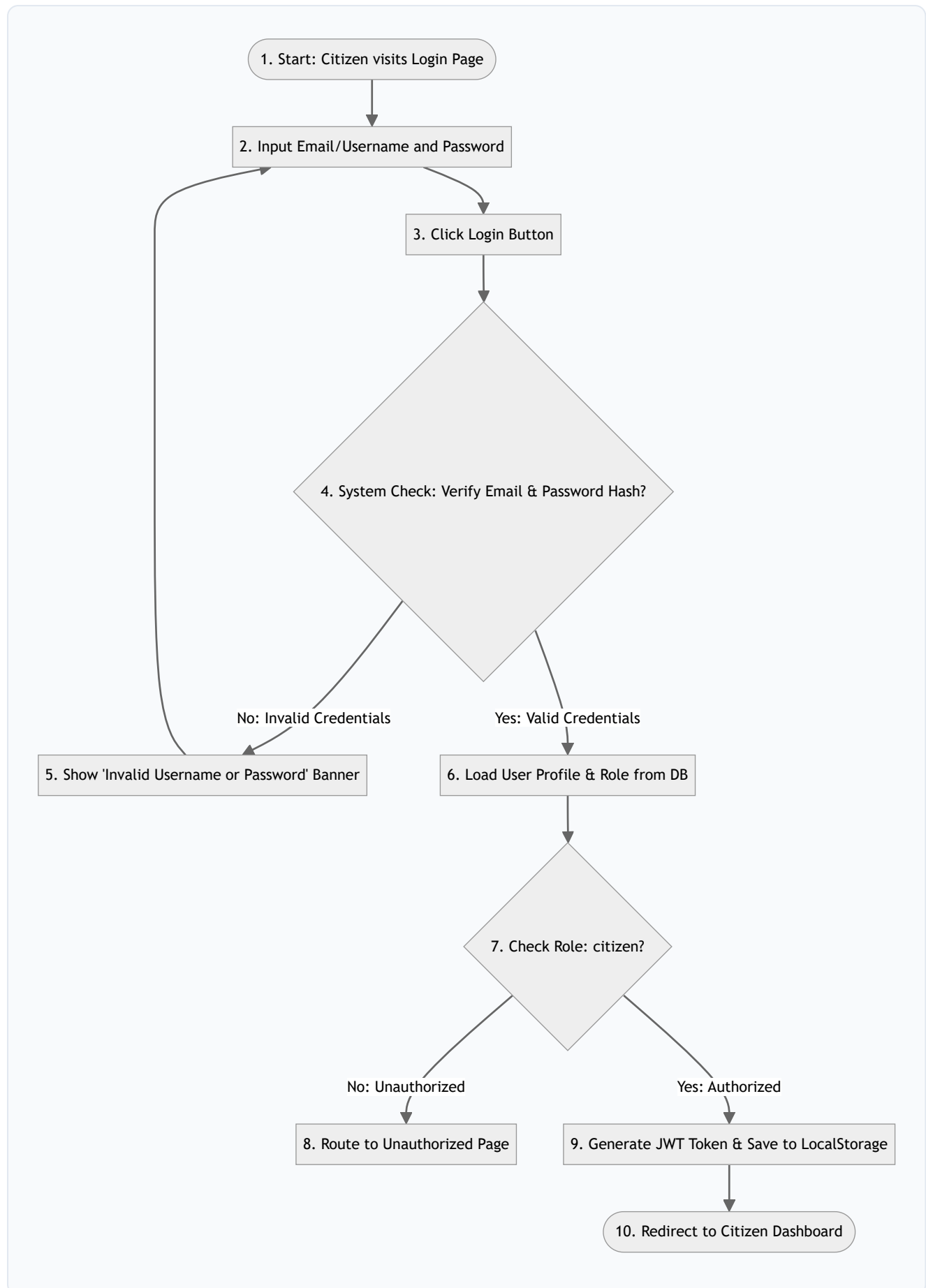
5. User Experience (UX) Flow Architecture

5.1 Flow 1: Citizen Registration



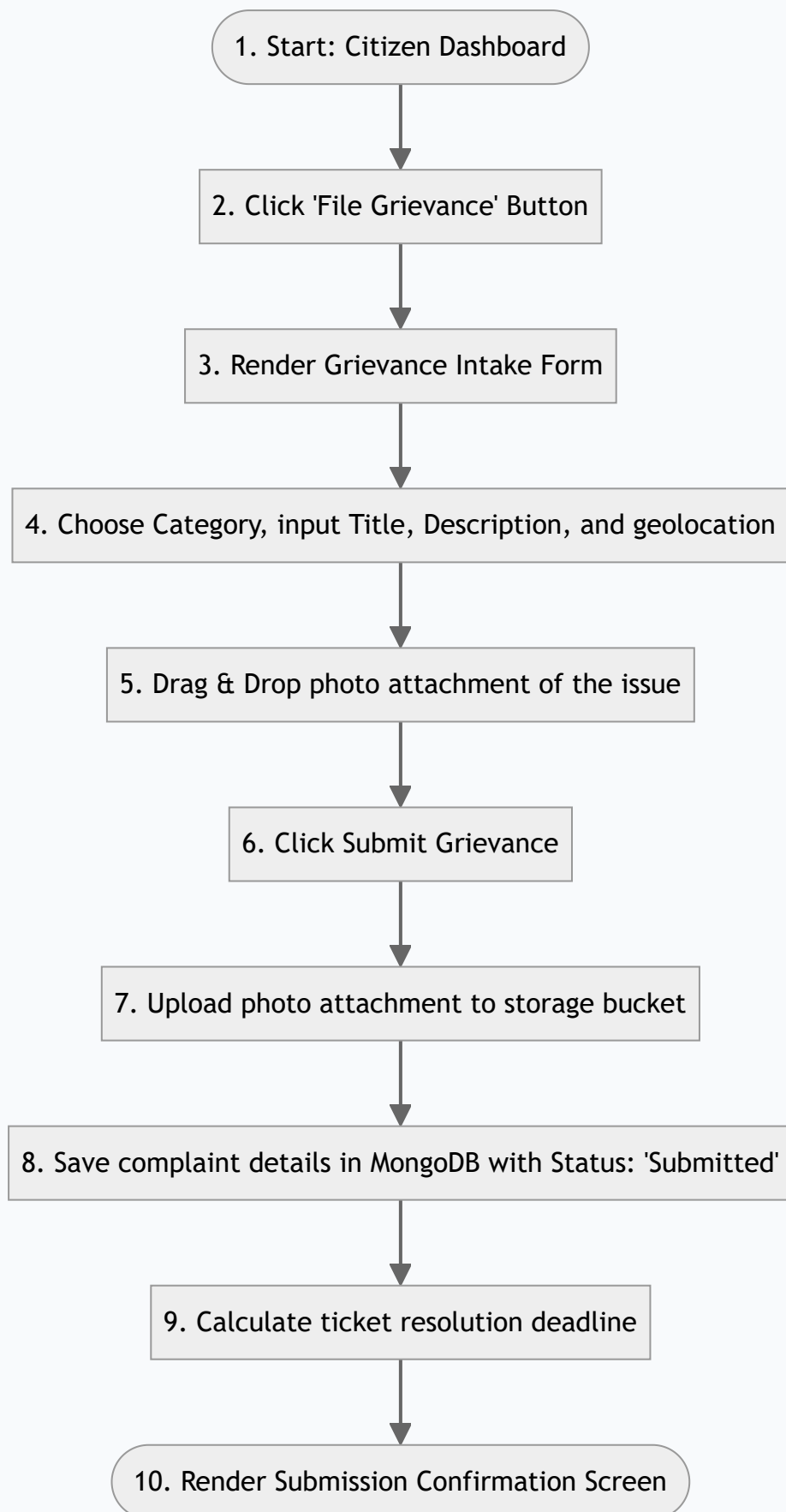
1. **Form Intake:** New citizens navigate to `CIT-00` and enter their personal details.
 2. **Format Verification:** The browser checks validation constraints (Aadhaar length, email patterns, password complexity) dynamically.
 3. **Authentication Verification:** The backend API verifies the uniqueness of the credentials against MongoDB records.
 4. **Security Hashing:** Passwords are encrypted with a 12-round bcrypt salt before saving.
 5. **Completion:** The citizen is redirected to the login screen `PUB-03`.
-

5.2 Flow 2: Citizen Login



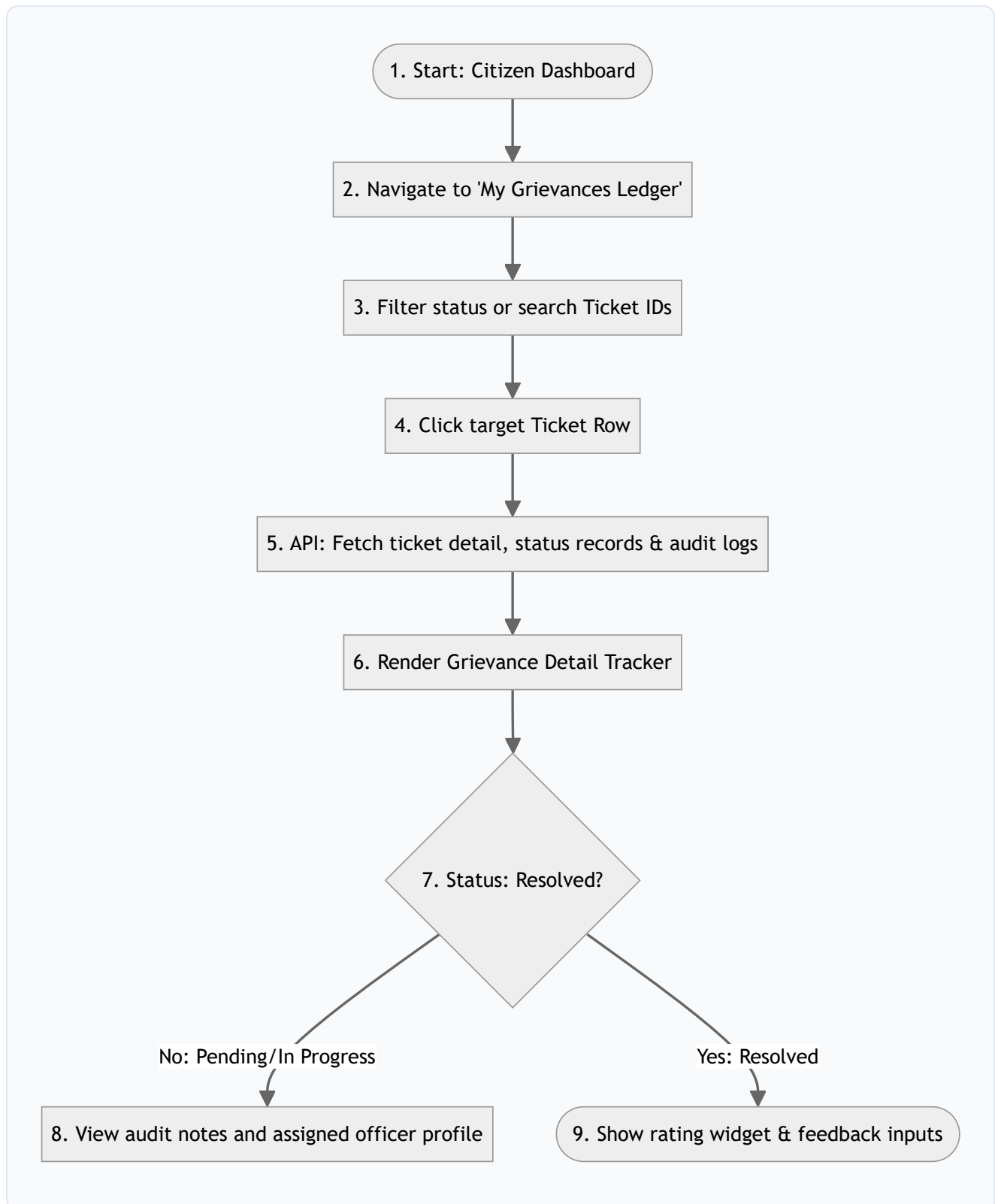
1. **Submission:** The citizen enters credentials on the login screen.
 2. **Authentication check:** The Node API verifies credentials using bcrypt hashing and loads the user role.
 3. **Token Storage:** A JWT token is generated and saved in `localStorage`.
 4. **Route Dispatch:** The system redirects the user to `/citizen/dashboard`.
-

5.3 Flow 3: Complaint Submission



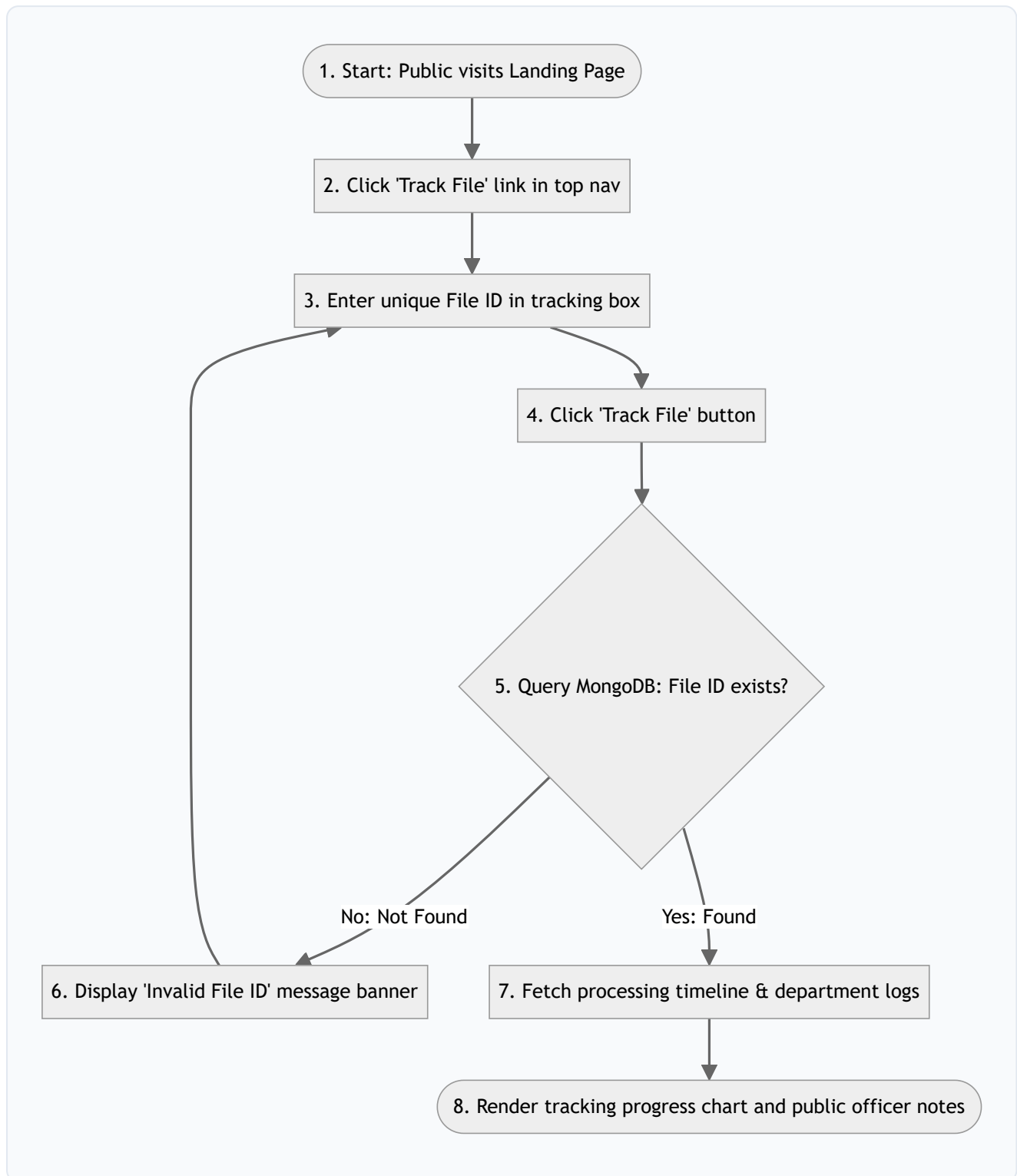
1. **Form Entry:** The citizen opens `CIT-02` and enters details (category, title, description, and location).
 2. **Attachment Upload:** The citizen uploads a JPG/PNG image, which is stored in S3 bucket storage.
 3. **Document Creation:** A POST query writes the complaint to the database, setting the default status to `"Submitted"`.
 4. **Deadline Assignment:** The system calculates the resolution deadline based on the category SLA rules.
 5. **Receipt:** The system displays a confirmation screen with the generated ticket ID.
-

5.4 Flow 4: Complaint Tracking



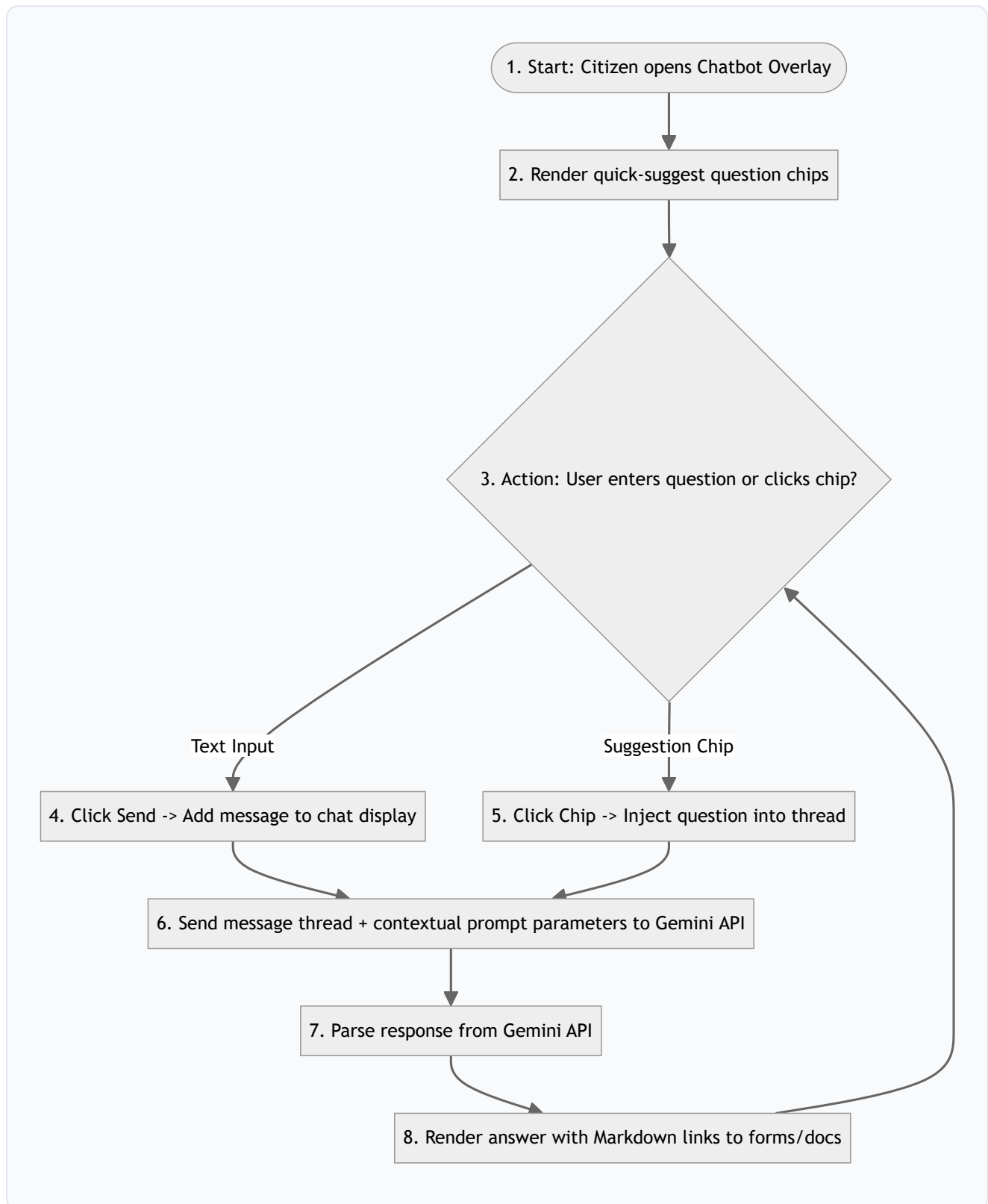
1. **List View:** The citizen opens `CIT-04` to view their registered complaints.
2. **Data Fetch:** An API request fetches the history of complaints matching the user's ID.
3. **Detail Tracker:** Clicking a ticket opens `CIT-05` to display the active status timeline and assigned officer details.

5.5 Flow 5: File Tracking (Universal Search)



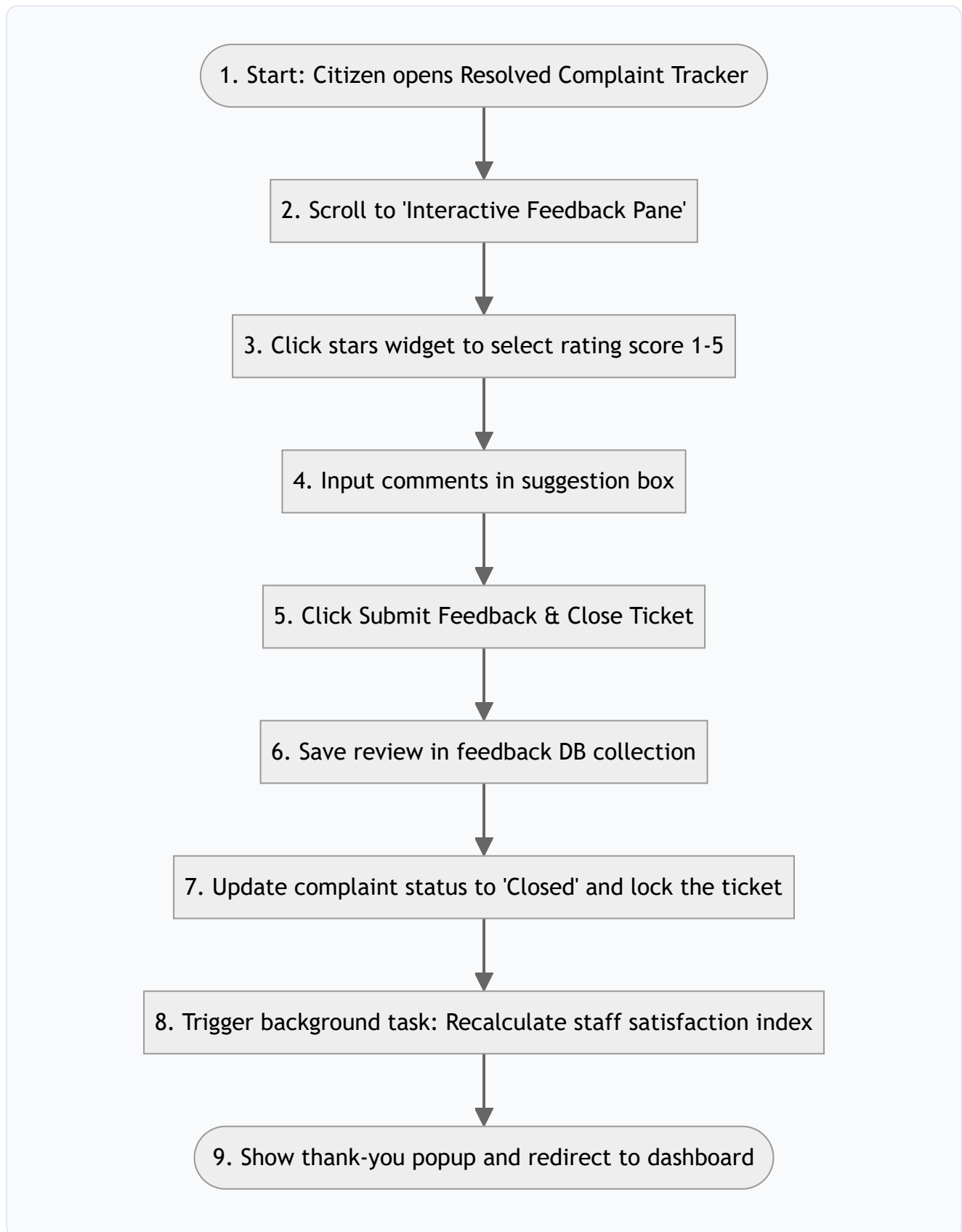
1. **Search Interface:** An anonymous user visits `PUB-02` and enters a unique File ID.
2. **Database Query:** The API queries the `fileTracking` collection for matches.
3. **Timeline Display:** The system displays the processing timeline and public officer comments.

5.6 Flow 6: Chatbot Usage



1. **Overlay Drawer:** The citizen opens the floating chat drawer on `CIT-06`.
2. **AI Query:** Suggestion chips provide quick FAQ links, or the user can type a question.
3. **Gemini AI Response:** The backend processes the input with a system prompt and fetches a response from the Gemini API.
4. **Render Response:** The response renders with markdown links to guide the user.

5.7 Flow 7: Feedback Submission

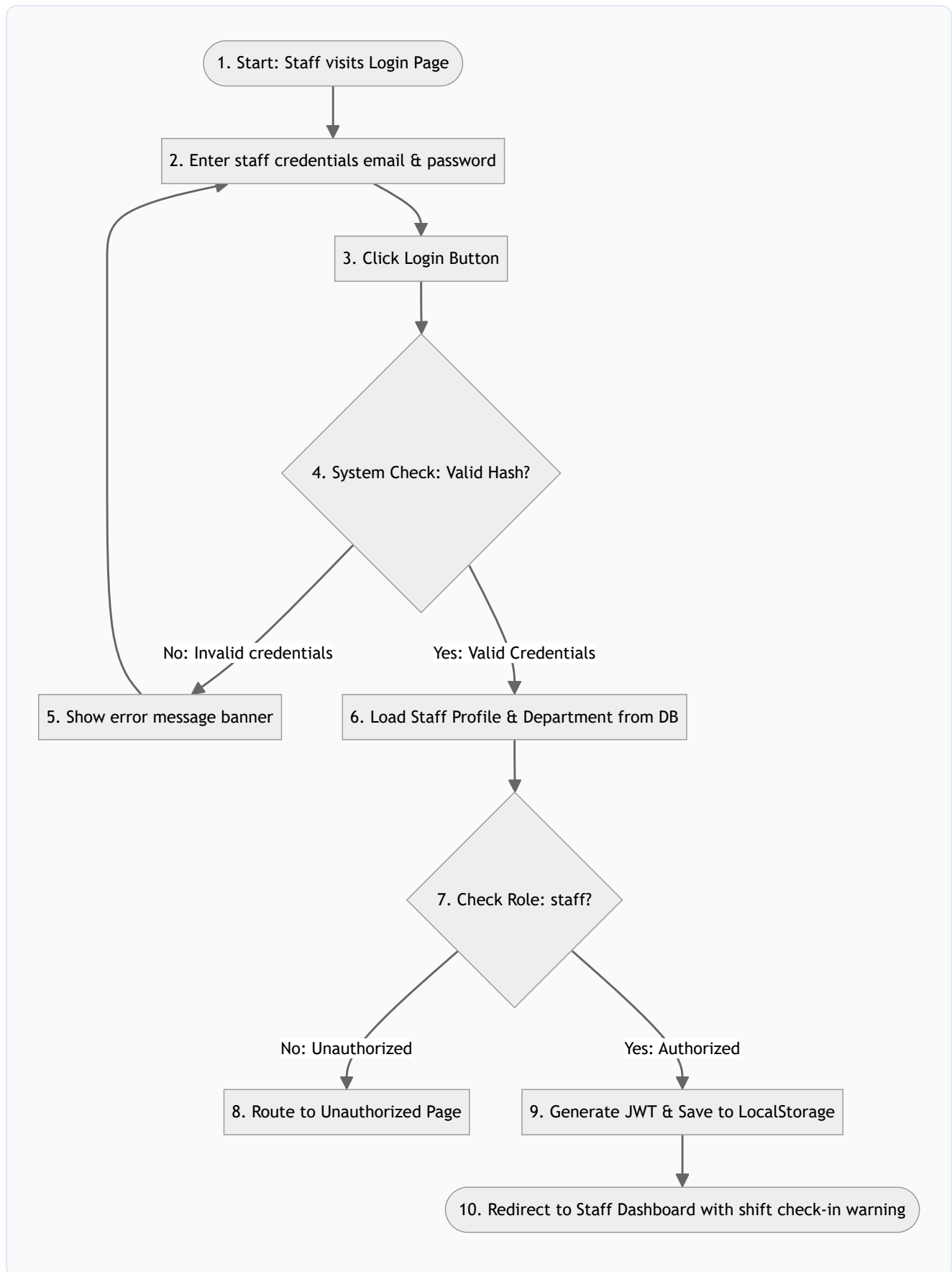


1. **Feedback Panel:** The citizen opens a ticket marked "Resolved" on `CIT-05`.

2. **Submit Rating:** The citizen rates the service and submits a review.

3. **Database Transaction:** The system saves the review in the `feedback` collection and updates the complaint status to `"Closed"`.
 4. **Recalculate Scores:** A background task recalculates satisfaction metrics for the assigned officer.
-

5.8 Flow 8: Staff Login

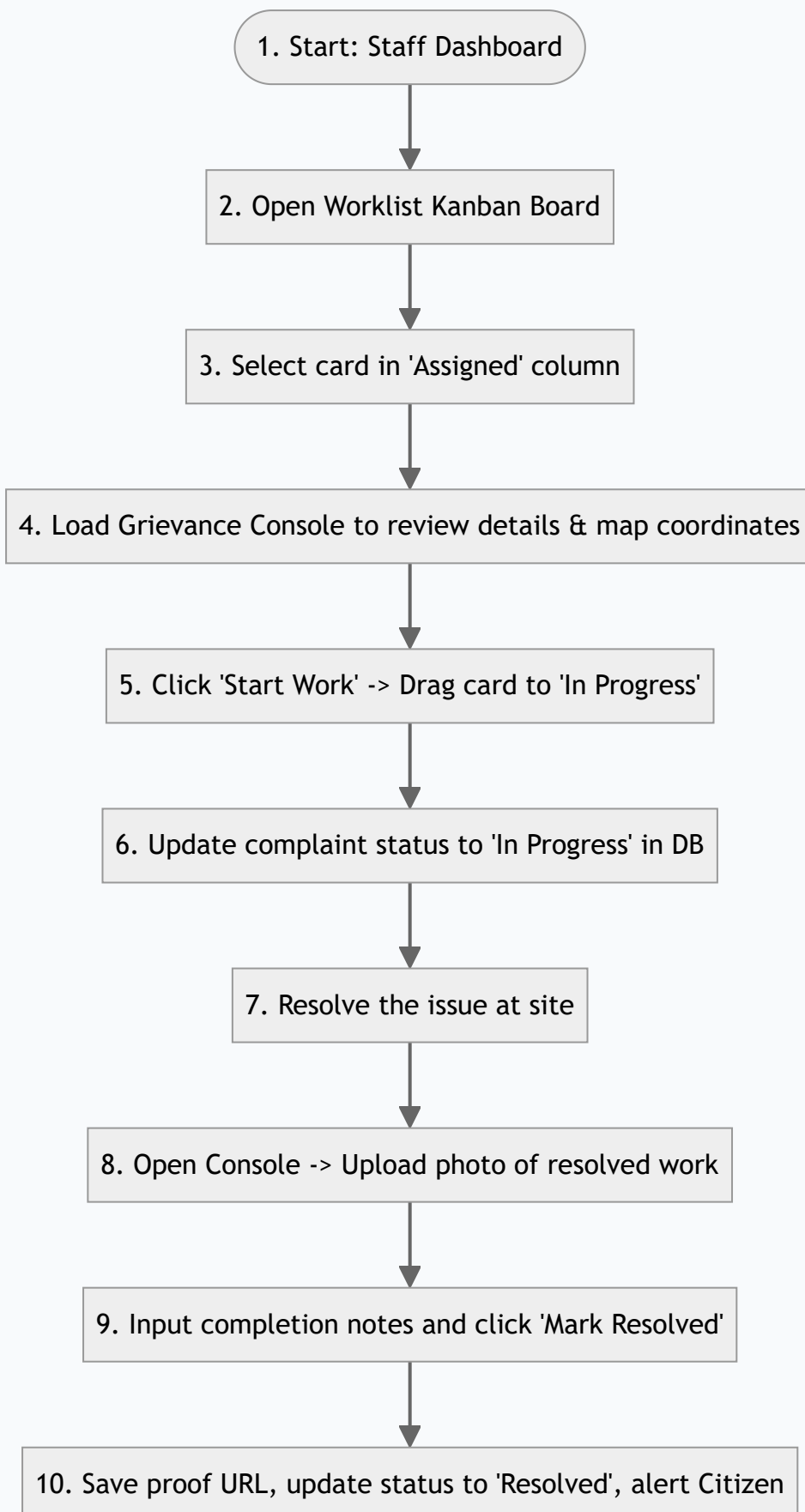


1. **Credentials Entry:** The staff member enters credentials on the login screen.

2. **Role Verification:** The backend verifies credentials and confirms the "staff" role.

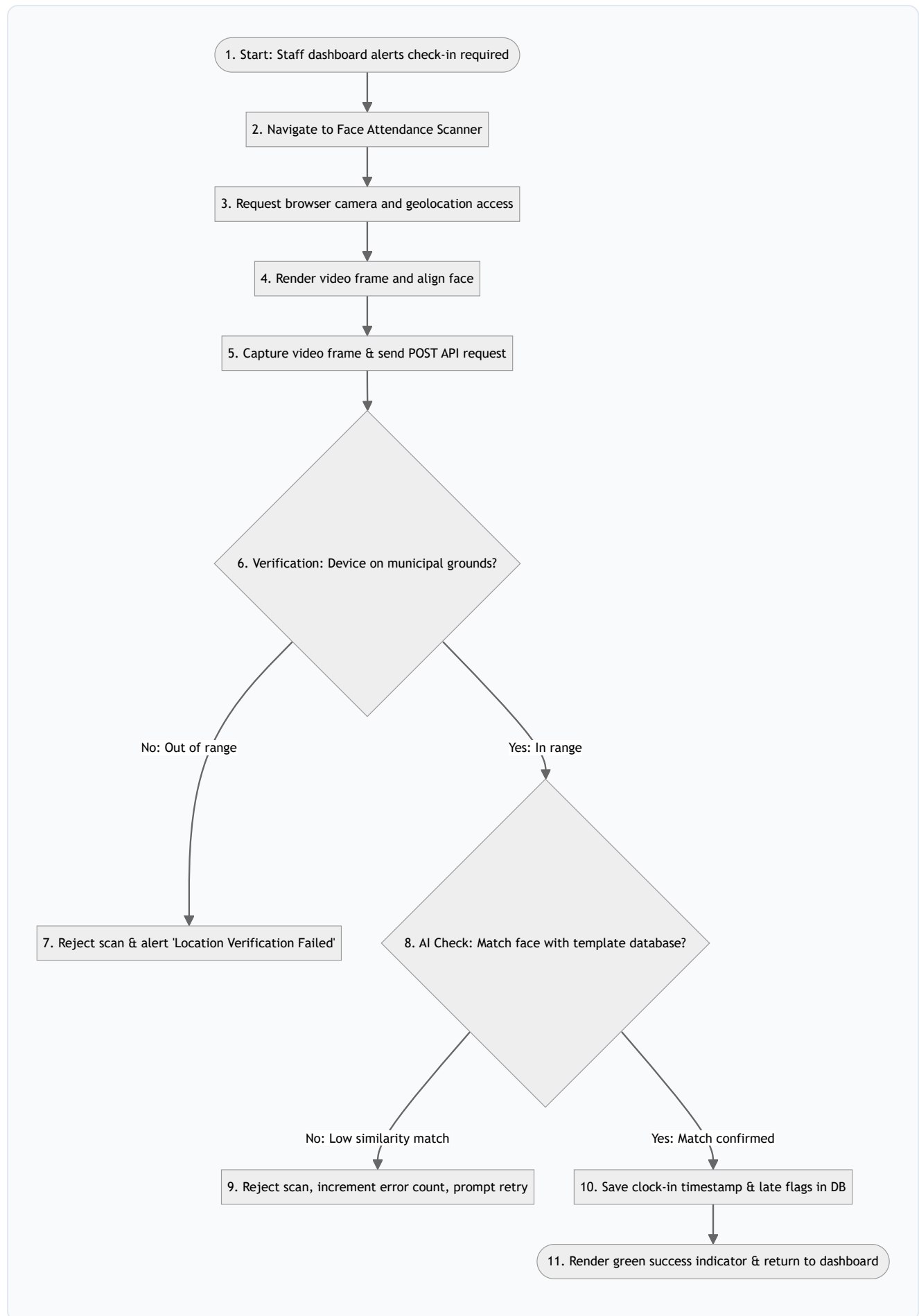
3. **Redirect:** The system saves the JWT and redirects the employee to STF-01 .

5.9 Flow 9: Staff Complaint Handling



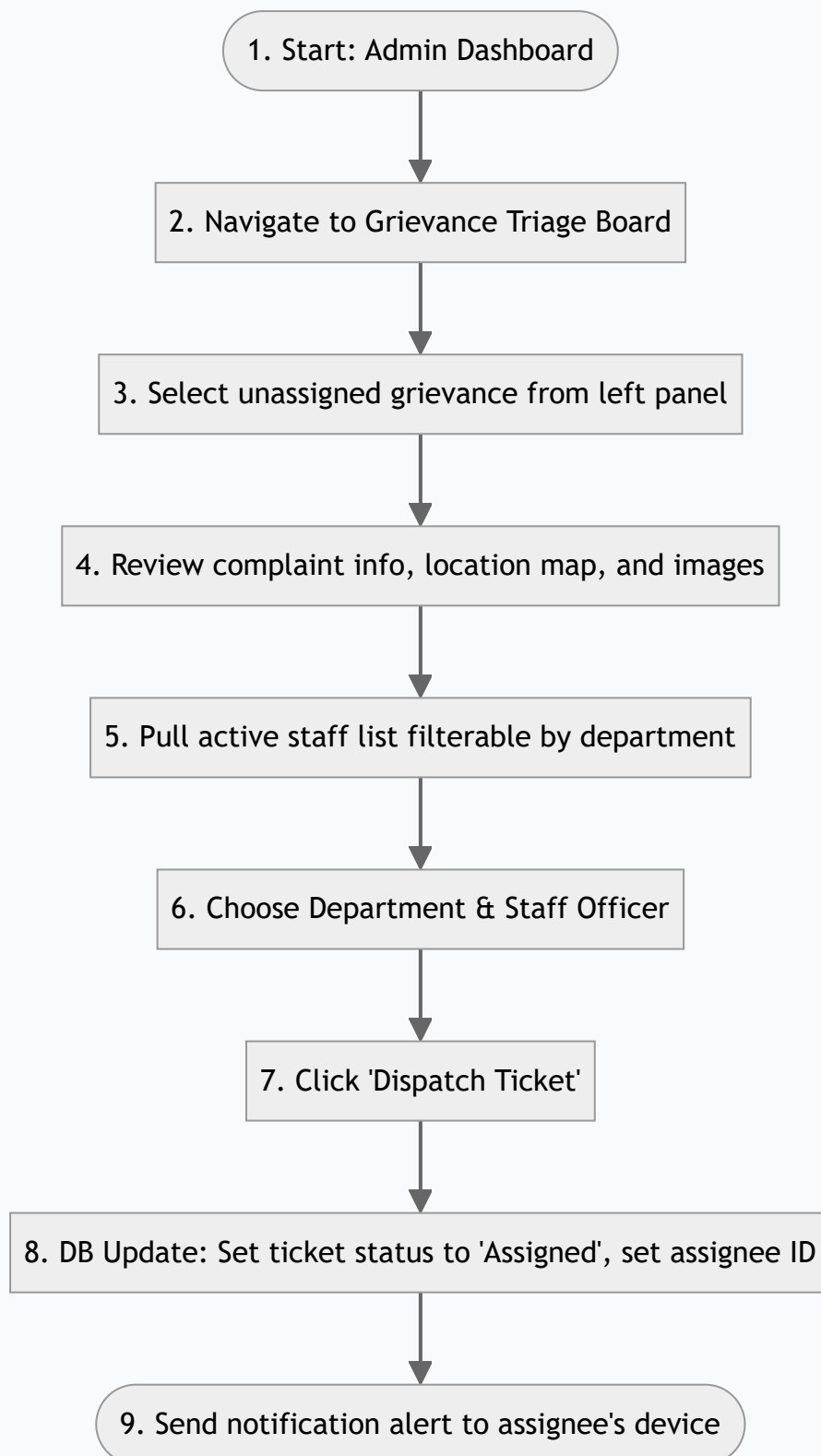
1. **Kanban Interaction:** The staff member opens their task board on `STF-03` .
 2. **Start Task:** Moving a card to "In Progress" updates the ticket status in the database.
 3. **Resolution Upload:** After completing work, the officer uploads a photo and comments.
 4. **Status Change:** The ticket status is set to "Resolved", and the citizen is notified.
-

5.10 Flow 10: Attendance Monitoring (Biometric Scanner)



1. **Start Scanner:** The employee navigates to the biometric scanner `STF-02`.
 2. **Hardware check:** The system requests webcam and GPS location permissions.
 3. **Frame Capture:** The webcam captures a video frame and sends it to the API.
 4. **DeepFace Verification:** The backend compares the scan against the employee's template.
 5. **Log Entry:** On confirmation, the system logs the clock-in timestamp and checks against the shift start time.
-

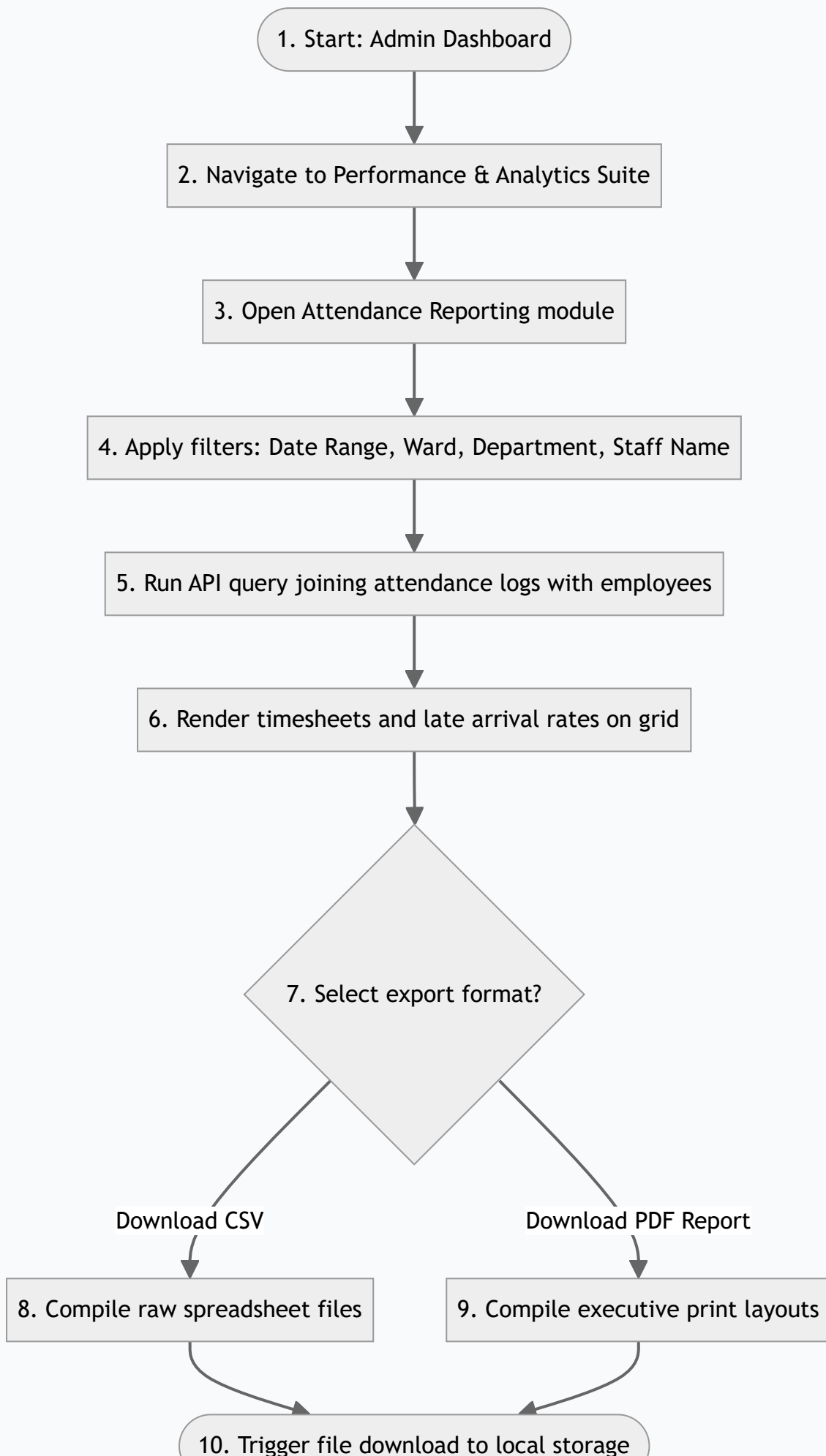
5.11 Flow 11: Admin Complaint Management (Triage & Dispatch)



1. **Triage Inspection:** The admin opens the triage board `ADM-03`.
2. **Review Ticket:** The admin selects an unassigned ticket and reviews details and location.
3. **Dispatch:** The admin selects a department and assigns the ticket to a staff member.

4. **Update Database:** The system updates the ticket status to "Assigned" and notifies the staff member.

5.12 Flow 12: Admin Attendance Reporting



-
1. **Reporting Interface:** The admin opens the reporting interface on `ADM-05` .
 2. **Filter Data:** The admin filters records by date range, department, or location.
 3. **Generate Report:** The system queries the `attendance` collection and compiles the timesheet data.
 4. **Download:** The admin downloads the report as a CSV or PDF file.

6. Low-Fidelity UI Wireframe Specifications

This section provides structural wireframes for key interface screens.

6.1 Citizen Portal Wireframes

6.1.1 Landing Page (PUB-01)

```
+-----+
| [Logo] Smart Municipal Portal                               File Tracker  Login  [Register] |
+-----+
| [ Banner Announcement Slider: Critical Water Line Repair Updates... (<) (>) ] |
+-----+
|
| Welcome to Citizen Assistance Hub
| Access municipal benefits, log grievances, and query chatbot assistance.
|
| +-----+ +-----+ +-----+ |
| | File Grievance      | | View active Schemes    | | Track File      | |
| | Report issues to officers| | Social welfare programs | | Check status   | |
| | [ Launch Form ]      | | [ Browse Directory ]   | | [ Track Now ]   | |
| +-----+ +-----+ +-----+ |
|
| Emergency Contacts Directory
| +-----+
| | Dept Name      | | Helpline Number | | Email Contacts      | |
| |-----|-----|-----|
| | Sanitation/Waste | | 1800-442-1209   | | waste-management@municipal.gov |
| | Water Supply     | | 1800-442-1215   | | water-board@municipal.gov    |
| +-----+
|
+-----+
| [?] Live AI Assistance Chat Bubble [Chatbot Launch Button]
+-----+
```

6.1.2 Login Page (PUB-03)

```
+-----+
| [Logo] Smart Municipal Portal                                     [Back to Landing] |
+-----+
|
|                                     +-----+
|                                     | Sign In Portal |
|                                     +-----+
|                                     | Email Address |
|                                     | [john.doe@email.com ] |
|                                     |
|                                     | Password |
|                                     | [*****] [o] |
|                                     |
|                                     | [] Remember credentials |
|                                     |
|                                     | [ Log In Button ] |
|                                     |
|                                     | Forgot password? |
|                                     | No account? [Register] |
|                                     +-----+
|
+-----+
```

6.1.3 Citizen Dashboard (CIT-01)

```
+-----+
| [Logo] Municipal Portal                                     Home  MyTickets InfoHub  [Profile] |
+-----+
| [Sidebar Menu] | Citizen Portal Dashboard |
| - Home | ----- |
| - Log Grievance | Welcome, John Doe! [ File Grievance ] |
| - My History |
| - Info Directory | +-----+ +-----+ +-----+ | | | | | |
| - Settings | | Active Tickets: 2 | | Resolved Issues: 8 | | Feedback: 4 |
| | +-----+ +-----+ +-----+ |
| |
| | Recent Activity Tracker | | | | | |
| | +-----+ |
| | | Ticket ID | Category | Status | Last Action | |
| | |-----|-----|-----|-----| |
| | | G-9812 | Potholes | In Progress | Assigned Staff | |
| | | G-9023 | Sanitation | Submitted | Awaiting Triage | |
| | +-----+ |
| | [ View All Activity History ] |
+-----+
| [?] Floating Assistant Chatbot |
+-----+
```

6.1.4 Chatbot Interface Overlay (CIT-06)

```

+-----+
| [Logo] Municipal Portal           Home  MyTickets InfoHub    [Profile] |
+-----+
| [Sidebar Menu] | Citizen Portal Dashboard                                [CHATBOT] |
| - Home         | -----+-----+
| - Log Grievance | Welcome, John Doe! | AI Helper[X]| | | | |
| - My History    | |-----|
| - Info Directory | +-----+ | Quick Links:|
| - Settings      | | Active Tickets: 2 | | [Apply Birth|
|                 | +-----+ | [Track File]|
|                 | |-----|
|                 | Recent Activity Tracker | AI: Welcome!|
|                 | +-----+ | How can I  |
|                 | | Ticket ID | Category   | Status     | | assist you? |
|                 | +-----+ |
|                 | |-----|
|                 | | User: [   ] |
|                 | | [SendBtn]  |
+-----+

```

6.1.5 Complaint Intake Form (CIT-02)

```

+-----+
| [Logo] Municipal Portal           Home  MyTickets InfoHub    [Profile] |
+-----+
| [Sidebar Menu] | Log New Grievance |
| - Home         | -----+-----+
| - Log Grievance | Category Select  | |
| - My History    | [ Roads (Potholes, Damages) v] |
| - Info Directory | |-----|
| - Settings      | Grievance Description |
|                 | [ Please enter details about the street damage... ] |
|                 | |-----|
|                 | Attach Photo Proof  |
|                 | +-----+
|                 | | Drag and drop files here or [ Choose Image File ] | |
|                 | +-----+
|                 | |-----|
|                 | Location Geotagging |
|                 | [ Locate using my current GPS coordinates ] |
|                 | [ Latitude: 12.9716 ] [ Longitude: 77.5946 ] |
|                 | |-----|
|                 | [ Save & Submit Grievance ] | [ Cancel Form ] |
+-----+

```

6.1.6 Grievance Details Tracker (CIT-05)

```

+-----+
| [Logo] Municipal Portal                               Home  MyTickets InfoHub    [Profile] |
+-----+
| [Sidebar Menu] | Ticket Status Tracker: #G-9812 | | |
| - Home         | ----- |
| - Log Grievance | Category: Potholes    | Submitted: 11 June | Status: Active |
| - My History    | |
| - Info Directory | Progress Timeline: |
| - Settings      | (X) Submitted == (X) Assigned == ( ) In Progress == ( ) Resolved |
| |
| | Assigned Officer Info: | | | |
| | +-----+ |
| | | Name: Robert Smith (Sanitation Dept) | Ph: 9812-442-00 | |
| | +-----+ |
| | |
| | Grievance Details: |
| | Description: Large pothole causing traffic blockage. |
| | Image Proof: [ pothole_img_01.jpg (Thumbnail Preview) ] |
| | |
| | Officer Update History: |
| | - 11 June 10:00 AM: Dispatched to Ward 12 field units. |
| | |
| | [ Back to Ledger History ] |
+-----+

```

6.1.7 Universal File Tracker (PUB-02)

```

+-----+
| [Logo] Smart Municipal Portal                               [Back to Home] |
+-----+
|
|                                     +-----+
|                                     | Universal File Lookup |
|                                     |-----|
|                                     | Enter Unique File ID |
|                                     | [ F-2026-9812      ] |
|                                     | |
|                                     | [   Query Status   ] |
|                                     +-----+
|
| File Processing History  v
| +-----+
| | Date          | Department          | Status          | Officer Comments |
| |-----|-----|-----|-----|
| | 10 June       | Registrar Office    | Initial Triage  | File Received    |
| | 11 June       | Deputy Commission   | Approved Sign   | Certificate Issued|
| | +-----+
|
+-----+

```

6.1.8 Grievance Evaluation Rating Interface (CIT-05 Resolved State)


```
+-----+
| [Logo] Staff Portal | [Back to Dashboard] |
+-----+
|
|                                     +-----+
|                                     | Face Check-in Scanner | | |
|                                     |-----|
|                                     | /-----\ |
|                                     | / Webcam \ |
|                                     | / Viewport \ |
|                                     | | (Align Face) | |
|                                     | \-----/ |
|                                     | \-----/ |
|                                     | [ Start Clock-In Scan]|
|                                     +-----+
|
| Device Logs & Validation:
| - GPS Coordinates: 12.9716 N, 77.5946 E (Verified: Municipal Office)
| - Verification Status: Ready, align camera viewport.
|
+-----+
```

6.2.3 Assigned Complaints Kanban (STF-03)

```
+-----+
| [Logo] Staff Portal | [Status: Clocked-In (09:00)] | [Logout] |
+-----+
| [Sidebar Menu] | Grievance Operations Kanban Board |
| - Home | ----- |
| - Attendance | Filters: [ All Wards v] [ Sorting: Priority v ] |
| - My Assigned Tasks |
| - Ledger Logs | +-----+ +-----+ +-----+ | | | | | | |
| | | ASSIGNED (3) | | IN PROGRESS(2)| | RESOLVED (12) | |
| | |-----| |-----| |-----| |
| | | [#G-9812] | | [#G-9011] | | [#G-8802] | |
| | | Potholes | | Sanitation | | Water Leak | |
| | | Ward 12 [High]| | Ward 12 [Med] | | Ward 12 [Low] | |
| | | | | | | | |
| | | [#G-9923] | | [#G-9022] | | [ View All ] | |
| | | Waste Outage | | Streetlight | | | |
| | | +-----+ +-----+ +-----+ |
+-----+
```

6.3 Admin Portal Wireframes

6.3.1 Admin Central Super Dashboard (ADM-01)


```

+-----+
| [Logo] Admin Central Control           [Server: Active]           [Logout] |
+-----+
| [Sidebar Menu] | Grievance Dispatcher Board | | | | |
| - Dashboard Home | ----- |
| - Staff Console | Unassigned Tickets | Complaint Details & Triage |
| - Grievance Triage | +-----+ |
| - Global Ledger | | G-9812: Pothole | Ticket: #G-9812 | Priority: [High] | |
| - Reporting Suite | | Ward 12 - 11 June | Desc: Deep pothole at 4th Lane. | |
| - System Config | | | Geotag: Ward 12 (Latitude: 12.97) | |
| | | G-9923: Water Leak | Attachment: [ pothole.jpg (Preview) ] | |
| | | Ward 12 - 11 June | | |
| | | | Assign Department: | |
| | | G-9991: Garbage | [ Sanitation Dept v ] | |
| | | Ward 03 - 10 June | Assign Officer: | |
| | | | [ Robert Smith (Workload: 2) v ] | |
| | | | [ Dispatch Grievance Ticket ] | |
| | | +-----+ |
+-----+

```

6.3.4 Attendance Timesheet Ledger (ADM-05)

```

+-----+
| [Logo] Admin Central Control           [Server: Active]           [Logout] |
+-----+
| [Sidebar Menu] | Attendance Report Auditor | | | | | | |
| - Dashboard Home | ----- |
| - Staff Console | Filters: [ 11 June ] [ Sanitation v ] [ Export PDF/CSV ] |
| - Grievance Triage | |
| - Global Ledger | Today's Attendance Logs Ledger: |
| - Reporting Suite | +-----+ |
| - System Config | | Name | Clock In | Clock Out | Coordinates | Status | |
| | |-----|-----|-----|-----|-----| |
| | | Robert | 08:55 AM | --:-- | 12.971, 77. | [Present] | |
| | | Sarah | 09:15 AM | --:-- | 12.971, 77. | [Late ] | |
| | | David | --:-- | --:-- | --:-- | [Absent ] | |
| | | +-----+ |
+-----+

```

6.3.5 Performance Charts & Analytics (ADM-05)

```
+-----+
| [Logo] Admin Central Control           [Server: Active]           [Logout] |
+-----+
| [Sidebar Menu] | Performance & Analytics Suite | | | | |
| - Dashboard Home | ----- |
| - Staff Console | Range: [ Last 30 Days v]           [ Run Custom Audit ] |
| - Grievance Triage | |
| - Global Ledger | +-----+ |
| - Reporting Suite | | Monthly Complaint Volume Trend (Line Chart) | |
| - System Config | | |
| | | Vol | | |
| | | 50 | | | /\ |
| | | 25 | | | /\ / \ |
| | | 0 | | | / \ / \ |
| | | +-----+ | |
| | | 1 7 14 21 28 (Days) | |
| | | +-----+ | |
| | | +-----+ +-----+ | |
| | | Avg Resolution Speed (Days) | | Citizen Satisfaction % | |
| | | Roads [=== 2.4 days] | | Sanitation [ 4.8 / 5.0 ] | |
| | | Sanitation [= 1.1 days] | | Streetlight[ 4.2 / 5.0 ] | |
| | | +-----+ +-----+ | |
+-----+
```

7. Reusable React Component Catalog

This section details the reusable React components designed to build the portals.

7.1 Component Prop & Customization Directory

Component Name	Purpose	Key Props	Reusability details	Screens Used
Button	Base UI trigger elements.	children , variant , size , isLoading , onClick	Supports variants: primary, secondary, danger, outline, ghost.	All screens.
Card	Content and layout blocks container.	title , children , hoverEffect , borderStatus	Standardized padding and shadow borders.	PUB-01 , CIT-01 , STF-01 , STF-03 , ADM-01 .
Modal	Overlay dialog panel.	isOpen , onClose , title , children	Renders using React Portal to prevent alignment issues.	CIT-05 , STF-04 , ADM-02 .
Table	Data grid displaying registry records.	columns , data , paginated , rowsPerPage	Handles paginated databases and sorting.	PUB-01 , CIT-04 , STF-05 , ADM-02 , ADM-04 .
Navbar	Header navigation component.	portal , user , attendanceStatus	Renders options based on the active role.	Main Shell Wrapper.
Sidebar	Navigation side menu.	role , collapsed	Collapsible sidebar layout with active states.	Portal Shell Layout.
SearchBar	Search input field.	placeholder , onSearch , debounceMs	Emits debounced value changes.	PUB-02 , CIT-04 , STF-03 , ADM-02 , ADM-04 .
NotificationPanel	Toast notification alert lists.	alerts , onClear	Render stack animations.	CIT-01 , STF-01 , ADM-01 .

Component Name	Purpose	Key Props	Reusability details	Screens Used
ChatWindow	Gemini chatbot overlay panel.	contextId , initialGreeting	Embedded widget supporting FAQ search prompts.	Floating overlay bubble.
StatusBadge	Labels indicating state status.	label , statusType	Maps status to colors (emerald, amber, crimson).	PUB-02 , CIT-04 , STF-03 , ADM-04 .
DashboardWidget	KPI metrics dashboard summary cards.	label , value , icon , colorScheme	Renders numbers and stats in grid blocks.	CIT-01 , STF-01 , ADM-01 .
WebcamCapture	Web camera module container.	onCaptureFrame , isActive , overlayTarget	Controls webcam API connections.	STF-02 , ADM-02 .
TimelineProgress	Stepper timeline components.	steps , activeStepIndex	Horizontal node timeline indicator.	PUB-02 , CIT-03 , CIT-05 .
AnalyticsChart	Graphical charting panel component.	type , datasets , labels	Integrates with Recharts for admin data.	ADM-01 , ADM-05 .

8. Figma Design System Tokens & Style Guide

A unified spatial and visual design system built on an 8-point grid.

8.1 Visual Style Variables

Color Palette System:

- Brand Primary (Deep Navy) : hsl(218, 51%, 14%) | #121e36
- Accent Secondary (Indigo) : hsl(244, 97%, 64%) | #5a4efc
- Neutral Background (Light Gray) : hsl(210, 15%, 97%) | #f8f9fa

Status Indicators:

- Success (Emerald Green) : hsl(142, 70%, 45%) | #22c55e
- Warning (Gold) : hsl(38, 92%, 50%) | #eab308
- Danger (Crimson Red) : hsl(0, 84%, 60%) | #ef4444

8.2 Typography Scale System

- **Header Typeface:** Outfit (sans-serif)
- **Body & UI Typeface:** Inter (sans-serif)
- **Display Header:** 32px (Bold, Line Height: 120%)
- **Page Title:** 24px (SemiBold, Line Height: 130%)
- **Card Header:** 18px (Medium, Line Height: 140%)
- **Body Text:** 14px (Regular, Line Height: 150%)
- **UI Details / Caption:** 12px (Regular, Line Height: 140%)

8.3 Spacing Rules

All layouts are configured on a strict 8-point spatial system:

- space-xxs (4px): Small label margins.
- space-xs (8px): Icon-to-text separation.
- space-sm (12px): Button interior padding.
- space-md (16px): Standard margins and gutters.
- space-lg (24px): Card interior margins.
- space-xl (32px): Dashboard section margins.
- space-xxl (64px): Screen boundaries spacing.

IMPORTANT

BIOMETRIC PRIVACY & LEGAL COMPLIANCE:

Biometric employee profiles (facial vector datasets generated using DeepFace) must be encrypted and stored in MongoDB using secure hashing protocols. The system must not store raw images on municipal databases after verification dataset registration is complete. All database connections and user portals must implement HTTPS protocols.

Section 9: Additional Frontend Architecture Enhancements

9.1 User Journey Maps

Citizen: Home → Login/Register → File Complaint → Track Complaint → Feedback.
Staff: Login → Attendance → Assigned Complaints → Resolution.
Admin: Login → Dashboard → Assign Complaints → Reports.

9.2 Route Mapping Table

/, /login, /register, /track, /citizen/dashboard, /citizen/complaint, /staff/attendance, /staff/tasks, /admin/dashboard, /admin/reports

9.3 Responsive Design Strategy

Mobile: Hamburger menu and single-column forms.
Tablet: Collapsible sidebar.
Desktop: Full dashboard layout and analytics panels.

9.4 State Management Architecture

AuthContext manages authentication and roles.
ComplaintContext manages complaint data.
ThemeContext manages appearance preferences.

9.5 Page ↔ API Mapping

Login → POST /api/auth/login
Complaint → POST /api/complaints
My Complaints → GET /api/complaints/my
Attendance → POST /api/attendance
Reports → GET /api/reports

9.6 Screen Responsibility Matrix

Complaint Page: Frontend + Backend
Chatbot: Frontend + Backend + AI
Attendance: Frontend + Backend + AI
Dashboard: Frontend + Backend
Reports: Frontend + Backend